



$$V_L = \frac{\sqrt{3}}{2\sqrt{2}} V_{dc} = \frac{\sqrt{3}}{2\sqrt{2}} \times 700 = 428.66 \text{ V} \quad (\text{Assump } V_{dc} = 700 \text{ in the worst case})$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = 247.49 \text{ V}$$

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{3 \times 2} = 1000 \text{ rpm}$$

$$s = \frac{N_r - N_s}{N_s} = 0.22$$

$$Z_T = R_s + \frac{R_r}{s} + jX_{eq} = 155.00 \angle 3.51^\circ$$

$$I_r = \frac{V_{ph}}{Z_T} = 1.5967 \angle -3.51^\circ$$

$$P_{mech} = 3 \cdot I_r^2 R_r \cdot \left(\frac{1-s}{s}\right) = 1.124 \text{ kW}$$

$$P_{loss} = 3 \cdot I_r^2 (R_s + R_r) = 61.19 \text{ kW}$$

$$\Rightarrow P_{DC} = P_{mech} - P_{loss} = 1.063 \text{ kW}$$

$$E_{580} = \frac{1}{2} C V^2 = \frac{1}{2} \times 10^4 \mu \times 580^2 = 1682 \text{ J}$$

$$E_{700} = \frac{1}{2} C V^2 = \frac{1}{2} \times 10^4 \mu \times 700^2 = 2450 \text{ J}$$

$$\text{therefore, } t_{ch} = \frac{E_{700} - E_{580}}{P_{DC}} = 722.5 \text{ ms}$$